

Week 3 Logic Test

The Professor's Riddles

Department of Science — Spotting Errors with AEIO Quantifiers

Name: _____

Date: _____

Whately's guidance: Before judging whether a claim is true, state its exact quantity and quality. Use the four standard forms: A = All S are P (universal affirmative), E = No S are P (universal negative), I = Some S are P (particular affirmative), O = Some S are not P (particular negative). "Most" is a strong particular claim and must be justified by evidence; it is not the same as "All." Overreaching with All/None when only Some holds, or claiming Most when evidence is unclear, produces false propositions even if the subject and predicate are right. Logic tests the structure of what is asserted.

The Puzzle Story

At the Academy of Straight Thinking, the Department of Science occupies the east wing. The professor has prepared four "riddles"—lists of statements about important scientific categories—for a classification activity. Each list is meant to contain only statements that could belong in a *true* description of that category.

The professor writes on the board:

"Tell me where my riddles go wrong. Some of these statements overreach. Use AEIO logic to find the errors in scope."

Below are the four riddles. For each category, decide which statements could belong in a true riddle (they are accurate in their claims about quantity) and which are false because they use All, None, or Most too broadly or incorrectly.

Part I: The Four Riddles

For each riddle, mark statements as T (could belong in a true riddle) or F (false due to overreach). Then, in the spaces provided after each riddle, list the false statement numbers and briefly explain the AEIO error (e.g., "claims All (A) but only Some (I) is true").

Riddle 1: Mammals

Decide which statements are true and which are false.

1. All are vertebrates.
2. Some lay eggs.

3. None can live in water.
4. Most have hair or fur at some point in life.
5. All give birth to live young.
6. Some can glide through the air.
7. None breathe through gills as adults.
8. Some are larger than cars.
9. All are warm-blooded.
10. Some use echolocation.

False statement numbers: _____

Explain the scope error(s) using AEIO terms for one or two false statements:

Riddle 2: Birds

Decide which statements are true and which are false.

1. All have feathers.
2. Some cannot fly.
3. None can swim.
4. All lay eggs.
5. Most can fly.
6. None have teeth as adults.
7. Some hunt at night.
8. All have hollow bones.
9. Some migrate thousands of miles.
10. None are warm-blooded.

False statement numbers: _____

Explain the scope error(s) using AEIO terms:

Riddle 3: Insects

Decide which statements are true and which are false.

1. All are invertebrates.
2. All have six legs as adults.
3. Some have wings.
4. None live in water.
5. Most are smaller than a human hand.
6. All undergo metamorphosis.
7. Some live in colonies.
8. None have antennae.
9. Some can sting.
10. All have exoskeletons.

False statement numbers: _____

Explain the scope error(s) using AEIO terms:

Riddle 4: Planets

Decide which statements are true and which are false.

1. All orbit a star.
2. None produce their own light.
3. Some have rings.
4. All are made of rock.
5. Some have no moons.
6. Most spin on an axis.
7. None have atmospheres.
8. Some are larger than Earth.
9. All are round because of gravity.
10. None can support storms.

False statement numbers: _____

Explain the scope error(s) using AEIO terms:

Part II: Spiral from Week 1

Week 1 emphasized asking what follows from what and distinguishing claims that can be tested. Choose one statement from any of the riddles above. Translate it into a precise AEIO proposition (name the form: A, E, I, or O). Then state what would follow if we accepted it as true.

Statement chosen (riddle and number): _____

Precise AEIO form: _____

Subject: _____

Predicate: _____

If we accept this precise proposition, what (if anything) logically follows about the category? Briefly explain.

Part III: Spiral from Week 2

Week 2 focused on fixing terms that shift meaning. Scan all four riddles. Are there any words that could be "two-faced" (carry different meanings in different statements or contexts)? If yes, identify one and fix it. If no clear cases, explain why the statements here keep terms steadier than the Week 2 examples.

Two-faced word found (or "none") : _____

Fixed meaning for this test: _____

Why fixing (or the absence of shifting) matters before judging scope:

Part IV: Repair One False Claim

Choose one false statement from the riddles (note which). Rewrite it as a true proposition using the correct AEIO quantifier. Then add one brief real-world observation that would support the repaired claim.

Riddle and statement number: _____

Repaired true version (with correct quantifier): _____

One supporting observation or fact: _____

Why does the precise AEIO version make the claim easier to test or defend?

Part V: What Logic Can Decide Here

For each item, mark whether **Logic (using AEIO)** can help test or expose the error or **Logic alone cannot settle this (needs facts outside logic)**.

1. Whether "All mammals give birth to live young" overclaims because platypuses lay eggs. _____
2. Whether the subject term "planets" is distributed in a universal claim. _____
3. Whether insects in fact live in water as adults or larvae. _____
4. Whether changing "All are warm-blooded" to "Some are warm-blooded" changes the quality of the proposition. _____
5. Whether birds are the only animals with feathers (definition or fact). _____
6. Whether "Most" claims can ever be strict A or E propositions. _____

Explain your marking for two of the items above:

Final Whately Reflection

In one or two sentences, explain why a scientist (or any thinker) must pay close attention to the exact words "all," "some," "none," and "most" before using a set of statements as a reliable riddle or description of a category.
